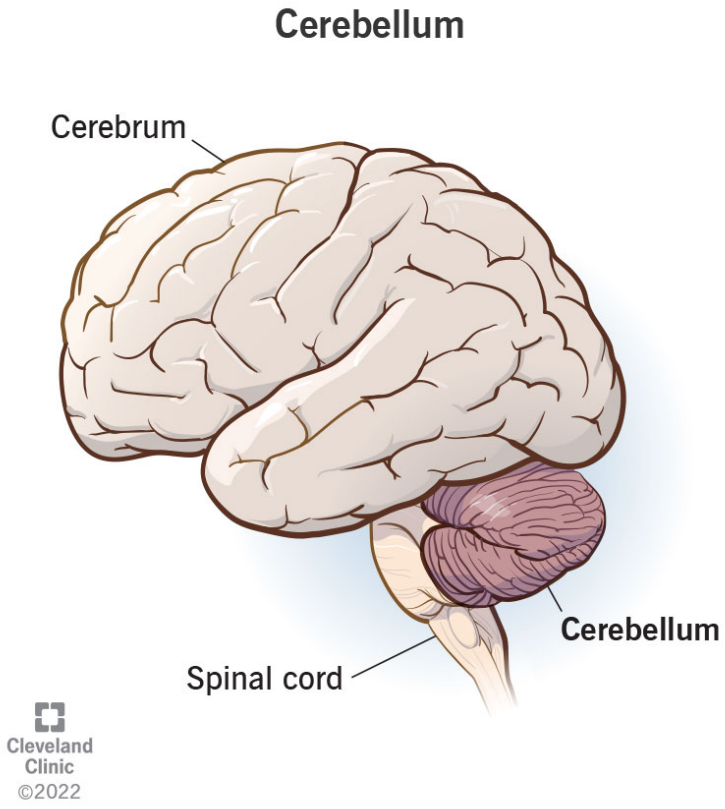
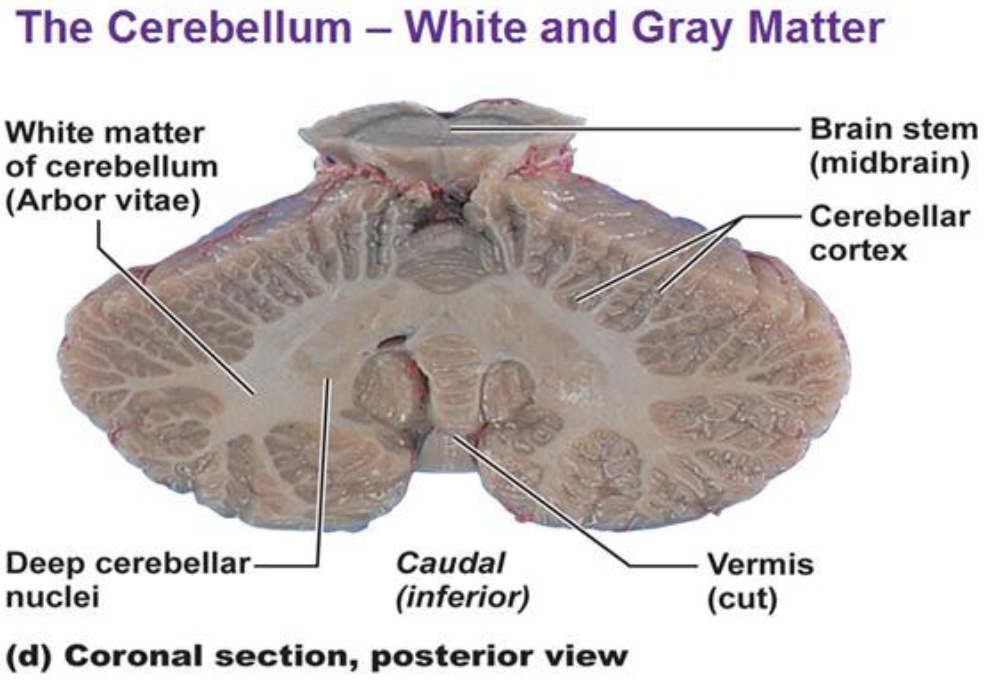
Neural circuits of the cerebellum and proteins in its synaptic vesicles

Cerebellum is a part of the brain located at the back of the head, just above and behind where spinal cord connects to the brain itself, just beneath the occipital and temporal lobes of the cerebrum and behind the brainstem. It makes up about 10% of total brain volume but holds more than half (50% to 80%) of the brain's total neurons The name “cerebellum” comes from Latin and means “little brain.



It is divided into two cerebellar hemispheres connected by a narrow midline strip called the **vermis**. The outer surface features a tightly folded sheet of gray matter called the **cerebellar cortex**, while the internal white matter branches out in a tree-like pattern known as the **arbor vitae** ("tree of life")



Primary Functions

- **Movement Coordination:** It times and adjusts muscle actions to produce smooth, fluid voluntary movements.
- **Balance and Posture:** It processes sensory information from the inner ear vestibular system and body receptors (proprioceptors) to maintain equilibrium.
- **Motor Learning:** It helps the body practice and fine-tune physical skills, such as riding a bicycle, playing an instrument, or hitting a ball.
- **Cognitive Roles:** Research shows it also contributes to non-motor functions, including language processing, attention, and emotional regulation

1. The major neural circuits of the cerebellum

The cerebellar cortex contains five major neurons: granule cells, Purkinje cells, basket cells, stellate cells, and Golgi cells. Climbing fibers and mossy fibers provide major incoming signals.

a) PURKINJE CELLS:

Biggest neurons in the body.

Have extensive dendritic arbors that extend throughout the molecular layer.

Cells receive the climbing fibres at the stem of dendritic tree.

Also receives axons of Stellate cell & basket cells.

Axons of these cells project to the deep cerebellar nuclei & few to the vestibular nuclei.

b) GRANULE CELLS:

Small, numerous, spherical neurons occupying greater part of the granular layer.

Axons go up the molecular layer & bifurcate like letter 'T' called the parallel fibre.

These fibres connect the Stellate cells, basket cells & Golgi cells.

Mossy fibres makes glomerular synapse with the dendrites of the granule cells

c) STELLATE CELLS:

Confined to the molecular layer.

Dendrites synapse with the parallel fibres while the axons synapse with the dendrites of Purkinje cell(near their origin).

d) BASKET CELLS:

Lie in the deeper part of the molecular layer.

Axons form networks around the cell bodies of Purkinje cells.

e) GOLGI CELLS:

Large stellate cells lying in the granular layer.

Dendrites which project into molecular layer receive inputs from the parallel fibres.

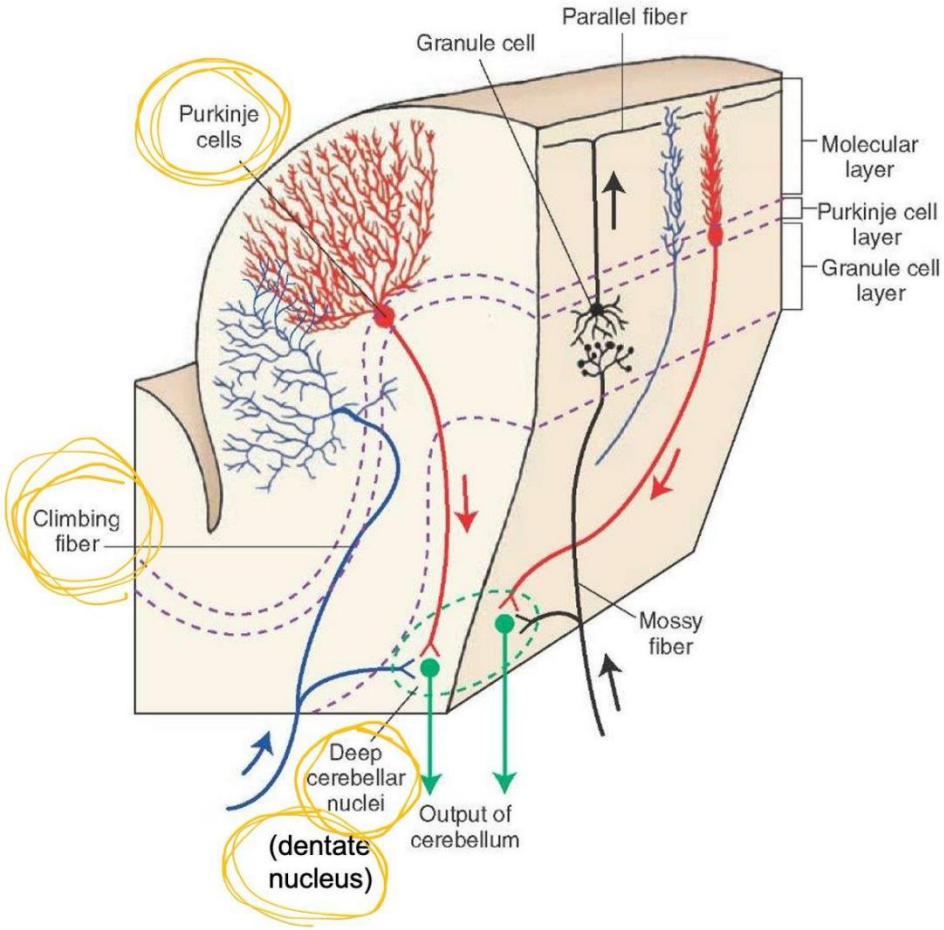
Golgi cells are responsible for feedback inhibition of granule cells.

The stellate cells, basket cells & Golgi cells act as interneurones & are inhibitory

Two primary fiber systems bring information into the cerebellum:

- **Mossy Fibers:** Originate from various brainstem and spinal cord nuclei. They send excitatory collaterals directly to the **Deep Cerebellar Nuclei (DCN)** and synapse on **granule cells** in the granular layer (releasing glutamate). Granule cells send their axons up to the molecular layer, splitting into T-shaped **parallel fibers** that excite Purkinje cells.

- **Climbing Fibers:** Originate exclusively from the **inferior olivary nucleus** in the medulla. They send excitatory collaterals to the DCN and wrap around the dendritic trees of **Purkinje cells**, creating a powerful "complex spike" that acts as an error or instruction signal for motor learning.



LAYERS OF THE CEREBELLAR CORTEX

a) Outer Molecular Layer:

Consists of Stellate cells & Basket cells

Innumerable synapses present executing a large amount of integrative work.

b) Intermediate/ middle layer of Purkinje Cells:

Have large flask shaped bodies with freely branching dendrites extending into the molecular layer & one axon passing to the cerebellar nuclei.

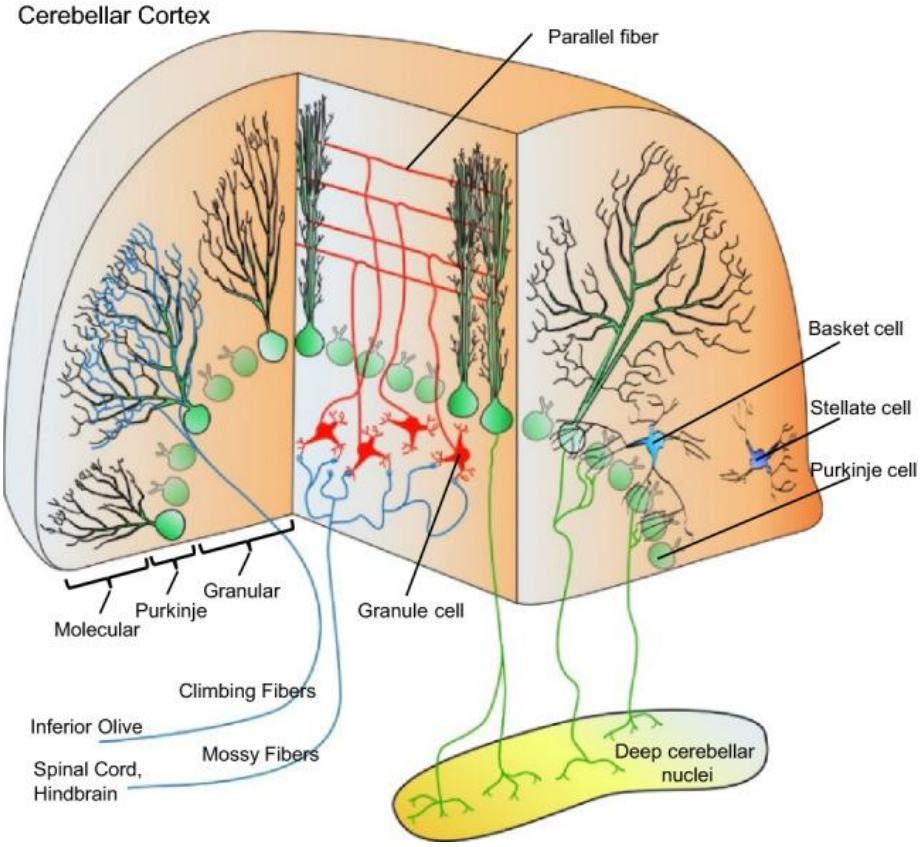
Contains cell bodies of Purkinje cells.

c) Inner Granular Layer:

Contains small Granule cells & Golgi cells.

Have multiple short dendrites & one long axon.

Axon ascends to the molecular layer & bifurcates into the transverse branches parallel to the surface



How the main circuit works

1. Mossy fiber pathway

Mossy fibers carry information from the spinal cord, vestibular system, pontine nuclei, and other brain regions. They excite granule cells, which send axons upward to form parallel fibers.

Main transmitter: glutamate (excitatory).

2. Climbing fiber pathway

Climbing fibers originate in the inferior olivary nucleus and form powerful excitatory synapses on Purkinje cell dendrites. They are important in motor learning and signaling movement errors.

Main transmitter: glutamate (excitatory).

3. Purkinje cell pathway

Purkinje cells integrate input from parallel fibers and climbing fibers. They are the principal output neurons of the cerebellar cortex and inhibit neurons in the deep cerebellar nuclei.

Main transmitter: GABA (inhibitory).

4. Inhibitory interneuron circuits

- Basket cells and stellate cells: inhibit Purkinje cells.
- Golgi cells: inhibit granule cells within cerebellar glomeruli.

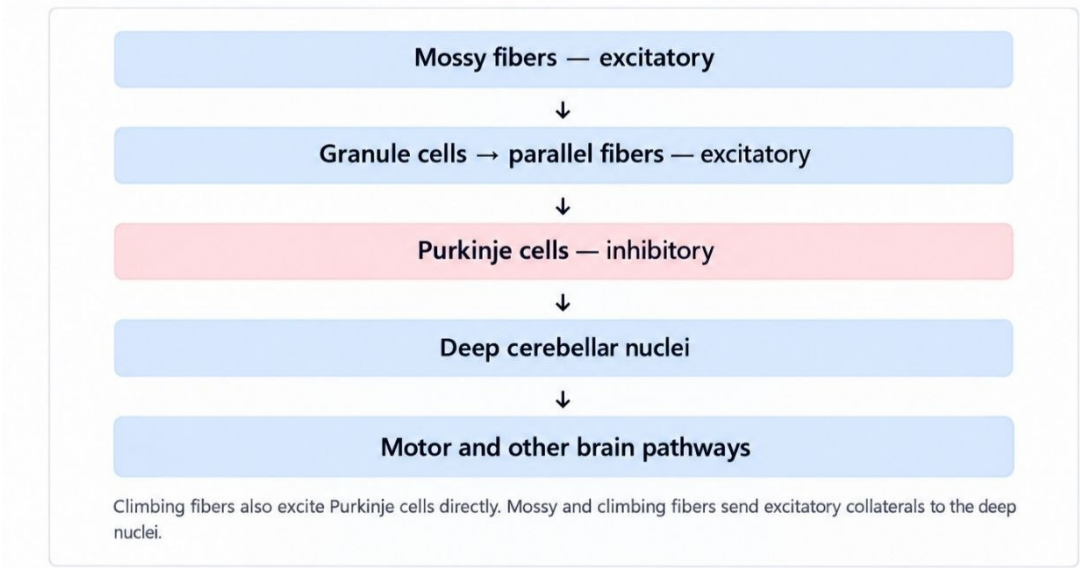
These circuits regulate the timing and strength of neural signals.

Main transmitter: GABA.

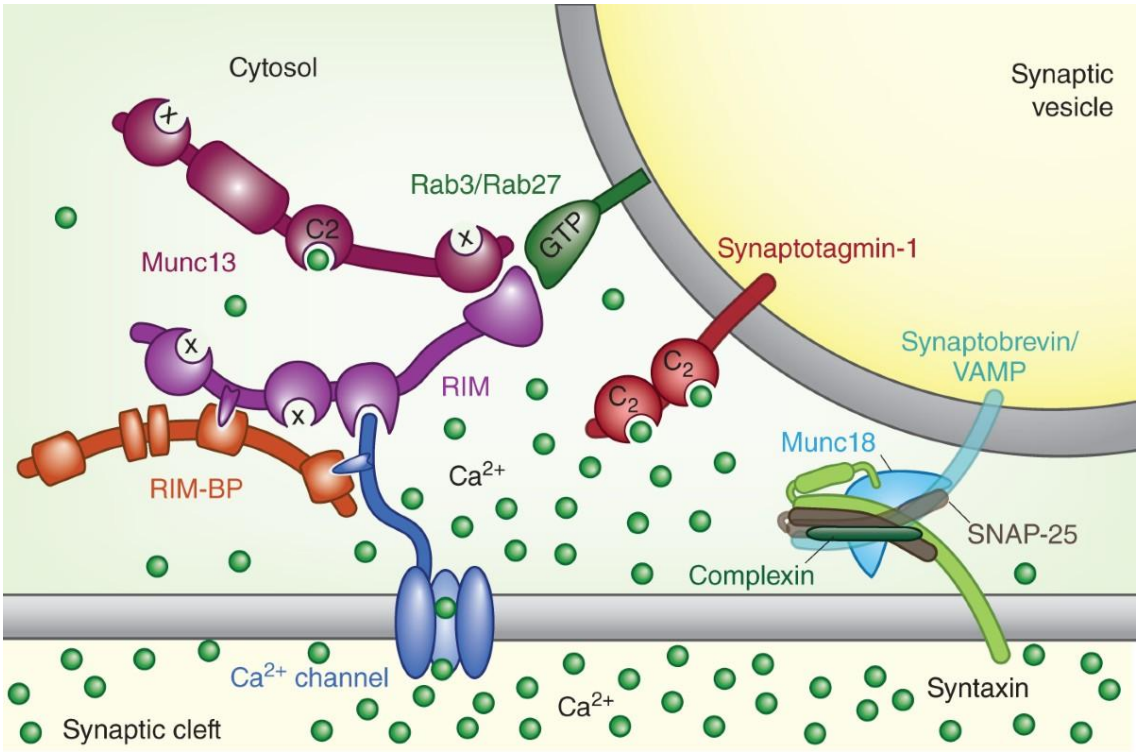
5. Deep cerebellar nuclei circuit

The dentate, interposed, and fastigial nuclei receive inhibitory input from Purkinje cells and excitatory input from collaterals of mossy and climbing fibers. Their output carries cerebellar signals to motor and other brain systems.

Many output neurons are glutamatergic and excitatory.



2. Proteins in cerebellar synaptic vesicles and their functions



Core Synaptic Vesicle Proteins & Functions

- **Synaptobrevin (VAMP-1 / VAMP-2)**
  - *Type:* Vesicular SNARE (v-SNARE) protein.
  - *Function:* Binds to plasma membrane t-SNAREs (**syntaxin** and **SNAP-25**) to form the tight SNARE complex that drives membrane fusion and exocytosis.

- **Vesicular Glutamate Transporters (VGLUTs, primarily VGLUT2 in mossy/climbing fibers)**
  - *Type:* Transporter protein.
  - *Function:* Uses the proton electrochemical gradient to actively pump glutamate from the cytoplasm into the vesicle lumen.

- **V-ATPase (Vesicular Proton Pump)**
  - *Type:* Multi-subunit enzyme pump (1–2 copies per vesicle).
- *Function:* Hydrolyzes ATP to pump H^+ ions into the vesicle, establishing the acidic and electrical gradient required for neurotransmitter loading.

- **Synaptotagmin**
  - *Type:* Integral membrane calcium sensor.
  - *Function:* Binds Ca^{2+} upon an action potential influx, removing clamp proteins and triggering rapid vesicle-to-plasma membrane fusion.

- **Synaptic Vesicle Glycoproteins (SV2A, SV2B, SV2C)**
  - *Type:* Multi-spanning transmembrane glycoproteins (SV2A is widely expressed in cerebellar gray matter).
  - *Function:* Regulates calcium-dependent vesicle exocytosis, interacts with synaptotagmin, and stabilizes vesicle pools.

- **Rab3A / Rab GTPases**
  - *Type:* Small GTP-binding protein.
  - *Function:* Cycles between GDP- and GTP-bound states to direct vesicle tethering and target-membrane docking at active zones.

- **Synapsins**
  - *Type:* Peripherally associated cytoskeletal-binding protein.
  - *Function:* Cross-links vesicles to form and maintain the **reserve pool** away from the active zone until phosphorylation triggers their release.

- **Munc13 Isoforms (e.g., Munc13-3)**
  - *Type:* Cerebellum-enriched priming protein.
  - *Function:* Essential for vesicle **priming** at parallel fiber–Purkinje cell synapses, regulating release probability and short-term synaptic plasticity.

2. Proteins and their functions

Protein	Main function
Synaptobrevin (VAMP1/VAMP2)	A vesicle SNARE protein that helps drive membrane fusion.
Vesicular glutamate transporters (VGLUTs)	Load glutamate into vesicles for excitatory neurotransmission.
V-ATPase	Uses ATP to pump protons into vesicles, creating the gradient needed for neurotransmitter loading.
Synaptotagmin	Acts as a calcium sensor that triggers rapid neurotransmitter release.
SV2A, SV2B, SV2C	Regulate vesicle release and interact with synaptotagmin.
Rab3A	Helps regulate vesicle trafficking and neurotransmitter release at synapses.
Synapsins	Help maintain the reserve pool of synaptic vesicles and regulate their mobilization.
Munc13 proteins, including Munc13-3	Support vesicle priming, preparing vesicles for release.

